# Legal Document Contradiction Detection using NLP and Machine Learning

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# ABSTRACT

This project presents an intelligent system for detecting contradictions in legal documents using Natural Language Processing (NLP) and Machine Learning techniques. Legal documents often contain complex and lengthy clauses, making manual identification of inconsistencies time-consuming and error-prone. The proposed system analyses pairs of legal statements—premise and hypothesis—to determine their relationship as entailment, contradiction, or neutral.

The model is trained on legal text datasets and uses text vectorization methods to capture semantic meaning. It automatically identifies conflicting statements and generates a structured output that highlights contradictions clearly for users. A user-friendly interface is developed using Streamlit, allowing users to input legal text and receive instant predictions.

This system improves the efficiency and accuracy of legal document analysis, reduces manual effort, and supports better decision-making in contract review and compliance processes.

# Chapter 1 - Introduction

## 1.1 Overview

In the legal domain, documents such as contracts, agreements, and policies are often lengthy and complex. These documents include multiple clauses, conditions, and obligations that must remain consistent throughout. However, due to their complexity, contradictions and inconsistencies can arise, leading to legal disputes and compliance issues.

The project "Legal Document Contradiction Summary Generator using Machine Learning and NLP" aims to automate the detection of contradictions in legal documents. The system analyses pairs of legal statements and identifies whether they are contradictory, entail each other, or are neutral.

## 1.2 Problem Definition:

Manual review of legal documents presents several challenges:

- Time-consuming and inefficient
- High chances of human error
- Difficult to detect hidden contradictions
- Not scalable for large datasets

**Problem Statement:**

To develop an automated system that detects and summarizes contradictions in legal documents using machine learning and NLP techniques.

## 1.3 Objectives:

The key objectives of this project are:

- To detect contradictions in legal text
- To apply NLP techniques for semantic understanding
- To build a machine learning classification model
- To generate structured summaries of conflicts
- To develop a Streamlit-based user interface

### 1.4 Motivation:

Legal professionals spend significant time reviewing documents to identify inconsistencies. Automating this process:

- Saves time
- Improves accuracy
- Reduces manual effort

This project is motivated by the need to apply AI in real-world legal applications.

### 1.5 Scope of the Project:

The scope of this project includes:

- Works on clause-level analysis
- Detects contradiction, entailment, neutral
- Uses ML + NLP techniques
- Includes Streamlit web application

Future scope:

- Full document analysis
- Integration with legal systems

### 1.6 Applications:

The developed system can be used in various real-world applications, including:

- Legal contract review
- Compliance verification
- Law firms' automation
- Document auditing systems

# Chapter 2: Literature Review

### 2.1 Introduction:

This chapter reviews existing research and methodologies related to contradiction detection and legal text analysis using Natural Language Processing (NLP) and Machine Learning (ML). It highlights various approaches used in text classification and semantic understanding.

### 2.2 Related Work:

Contradiction detection is a part of Natural Language Inference (NLI), where the relationship between two sentences is analysed. Researchers have used datasets such as SNLI, ANLI, and ContractNLI to train models for detecting entailment, contradiction, and neutral relationships.

Traditional machine learning models like Logistic Regression, Naive Bayes, and Support Vector Machines (SVM) have been widely used for text classification tasks. More recent approaches involve deep learning models such as LSTM and Transformer-based models like BERT, which provide higher accuracy.

### 2.3 NLP in Legal Text Analysis:

Machine Learning plays a crucial role in automatically analysing and classifying textual data. It enables systems to learn patterns from large datasets and make predictions on new, unseen data.

In text classification, the process typically involves:

- Preprocessing the text data
- Converting text into numerical features
- Training a classification model
- Evaluating performance

Machine learning models are effective because they can generalize patterns and adapt to different types of textual content.

### 2.4 Machine Learning Models:

Machine learning models are used to classify relationships between legal statements. In this project:

- Text is converted using TF-IDF vectorization

A classification model predicts:

- Contradiction
- Entailment
- Neutral

Logistic Regression is commonly used due to:

- High efficiency
- Good performance on text data
- Low computational cost

### 2.6 Summary

This chapter discussed various approaches for text classification and contradiction detection. It shows that combining NLP preprocessing with machine learning models is effective for analysing legal text.

# Chapter 3 – System Analysis and Dataset

## 3.1 Dataset Description:

The dataset used in this project is based on Legal Natural Language Inference (NLI) datasets such as ContractNLI and SNLI. These datasets are specifically designed to analyse relationships between pairs of sentences.

Each data entry consists of:

- Premise: A legal statement or clause
- Hypothesis: A statement to compare with the premise
- Label: Relationship between the two

The dataset is used to train the model to classify relationships into three categories:

- Contradiction
- Entailment
- Neutral

The dataset contains multiple examples of legal text, making it suitable for training machine learning models for contradiction detection.

## 3.2 Variables:

| Type | Variable Name | Description |
|---|---|---|
| **Input** | Premise | Legal clause or statement |
| **Input** | Hypothesis | Statement to compare |
| **Output** | Label | Relationship between statements |

**Category Mapping:**

| Label | Category |
|---|---|
| 1 | Contradiction |
| 2 | Entailment |
| 3 | Neutral |

### 3.3 Exploratory Data Analysis (EDA):

Exploratory Data Analysis (EDA) is performed to understand the dataset and identify patterns.

Key Analysis Performed:

- Checked dataset structure and data types
- Verified class distribution
- Observed text length and patterns

Insights:

- The dataset is well-balanced across all categories
- News titles and descriptions vary in length
- Each category contains distinct keywords

Example Observations:

- Sports news includes words like match, team, win
- Business news includes market, company, stock
- Sci/Tech includes technology, software, internet

These patterns help the model learn differences between categories.

### 3.4 Data Preprocessing:

Data preprocessing is an important step to prepare text for machine learning.

Steps Performed:

- Lowercasing: Converted all text to lowercase
- Removing punctuation: Cleaned unwanted symbols
- Tokenization: Split text into words
- Combining Text: Title and description merged (if applied)

**Feature Extraction (Vectorization)**

The cleaned text is converted into numerical form using:

TF-IDF (Term Frequency – Inverse Document Frequency)

This technique:

- Assigns importance to words
- Reduces impact of common words
- Helps model understand meaningful patterns

**Train-Test Split**

- Training Data → Used to train the model
- Testing Data → Used to evaluate performance

## 3.5 Summary

This chapter discussed the dataset structure, variables, exploratory analysis, and preprocessing techniques used in the project. Proper data preparation and feature extraction are crucial for building an accurate machine learning model for contradiction detection in legal documents.

# Chapter 4 – System Design

### 4.1 Introduction

This chapter explains the design and structure of the Legal Document Contradiction Summary Generator system. It describes how different components of the system interact to process legal text and detect contradictions.

The system is designed to be efficient, accurate, and user-friendly. It follows a pipeline approach where input text is processed step-by-step to generate meaningful predictions.

### 4.2 System Architecture

The architecture of the system follows a step-by-step pipeline:

User Input (Premise + Hypothesis)
↓
Text Preprocessing (Lowercase, Cleaning)
↓
Feature Extraction (TF-IDF)
↓
Machine Learning Model (Classification Model)
↓
Predicted Category Output (Contradiction / Entailment / Neutral)
↓
Streamlit Interface (Display Output Result)

**(Figure 4.1: System Architecture)**

Explanation of Components:

1. **User Input:**
   The user enters a premise and hypothesis through the Streamlit interface.
2. **Text Preprocessing:**
   The input text is cleaned by converting it to lowercase and removing unnecessary characters.
3. **Feature Extraction:**
   The cleaned text is converted into numerical format using TF-IDF vectorization.

4. **Machine Learning Model:**
   The trained model processes the input and predicts the relationship between the statements.
5. **Output:**
   The result (Contradiction, Entailment, or Neutral) is displayed to the user.

### 4.3 Model Design:

The system uses a machine learning classification model to identify relationships between legal statements.

**Working Principle:**

- Input text is converted into numerical vectors
- The model analyses patterns in the data
- The relationship is predicted based on learned features

**Output Classes:**

- Contradiction
- Entailment
- Neutral

**Why This Model?**

- Efficient for text classification
- Works well with TF-IDF features
- Fast prediction and easy implementation

### 4.4 Performance Metrics

To evaluate the model, the following metrics are used:

| Metric | Description |
|---|---|
| Accuracy | Percentage of correct predictions |
| Precision | Correct positive predictions |
| Recall | Ability to find all relevant cases |
| F1 Score | Balance between precision and recall |

These metrics help measure how well the model performs on unseen data.

### 4.5 User Interface (Streamlit App)

A web application is developed using Streamlit to make the system interactive and easy to use.

**Features:**

- Input fields for premise and hypothesis
- Predict button
- Displays result instantly

**Advantages:**

- Simple and user-friendly
- Real-time prediction
- No technical knowledge required

### 4.6 Summary

This chapter described the overall system design, including architecture, data flow, model design, performance metrics, and user interface. The system is designed to efficiently process legal text and provide accurate predictions in a user-friendly manner.

# Chapter 5 – Implementation

### 5.1 Introduction:

This chapter explains the practical implementation of the Legal Document Contradiction Summary Generator. It covers the tools and libraries used, step-by-step model development, integration of components, and deployment of the application using Streamlit.

### 5.2 Libraries Used:

The following Python libraries are used in this project:

| Library | Purpose |
|---|---|
| Python | Programming language |
| Pandas | Data manipulation and analysis |
| NumPy | Numerical operations |
| Scikit-learn | Machine learning model and TF-IDF |
| Streamlit | Web application development |
| Joblib | Model saving and loading |

### 5.3 Model Implementation Steps:

**Step 1: Load Dataset**

- The dataset (ContractNLI / SNLI) is loaded using Pandas
- Data structure is checked for consistency

**Step 2: Data Preprocessing**

- Converted text to lowercase
- Removed punctuation and unwanted characters
- Combined premise and hypothesis
- Cleaned and prepared text for processing

**Step 3: Feature Extraction**

- TF-IDF Vectorizer is used
- Converts text into numerical form
- Helps model understand word importance

**Step 4: Model Training**

- Machine learning model is trained using training data
- The model learns relationships between text pairs

**Step 5: Model Evaluation**

- Model tested on unseen data
- Performance evaluated using accuracy and other metrics

**Step 6: Model Saving**

- The trained model is saved using Joblib
- Files used:
    - Legal_ANLI_model.joblib
    - legal_ANLI_model_vectorizer.joblib

**5.4 Code Integration:**

The trained model is integrated into a Streamlit web application.

**Working of the Application:**

- Model and vectorizer are loaded using joblib.load()
- User inputs premise and hypothesis
- Text is combined and vectorized
- Prediction is made using the model
- Output is displayed on screen

**Label Mapping:**

- 0 → Contradiction
- 1 → Entailment
- 2 → Neutral

**5.5 Deployment:**

The application is deployed using Streamlit.

**Application Details:**

- **App Name:** Legal ANLI Predictor
- **Framework:** Streamlit
- **Input:** Premise and Hypothesis
- **Output:** Relationship prediction

**Features:**

- User-friendly interface
- Real-time predictions
- Lightweight and fast

### 5.6 Summary:

This chapter described the complete implementation process, including data preprocessing, feature extraction, model training, evaluation, and deployment. The integration of machine learning with a Streamlit interface makes the system practical, efficient, and easy to use.

# Chapter 6 – Results and Discussion

**6.1 Introduction:**
This chapter presents the results obtained from the machine learning model and discusses its performance in detecting contradictions in legal text. It also highlights observations from the dataset and the effectiveness of the Streamlit application.

**6.2 Model Performance Metrics:**
The performance of the model is evaluated using standard classification metrics:

| Metric | Description |
|---|---|
| Accuracy | Percentage of correctly predicted relationships |
| Precision | Correct positive predictions |
| Recall | Ability to identify all relevant cases |
| F1 Score | Balance between precision and recall |

**Performance Analysis:**
- The model performs well on unseen test data
- It accurately classifies relationships between legal statements
- TF-IDF helps in identifying important words
- The model provides stable and reliable results

**6.3 Visualization and Analysis:**
To better understand the model and dataset, different analyses are performed:

1. Relationship Detection
    - The model successfully identifies contradictions where statements conflict
    - Entailment cases are accurately detected when meanings align
    - Neutral cases are identified when no clear relation exists

2. Text Pattern Analysis

Certain keywords strongly influence predictions:

- Contradiction**:** not, cannot, prohibited
- Entailment**:** must, required, shall
- Neutral**:** unrelated or vague statements

3. Model Behaviour
    - Performs best on clear and direct statements
    - Slight confusion may occur in complex legal language
    - Accuracy depends on input quality and clarity

**6.4 Streamlit Application Output:**

The Streamlit web application allows real-time testing of the model.
Working:

1. User enters premise and hypothesis
2. Clicks on "Predict" button
3. Model processes input and displays result

Sample Output:

- Input**:** "The contract allows termination at any time"
- Hypothesis**:** "The contract cannot be terminated"
- Output**:** Contradiction

**Advantages of the Application:**

- Easy to use interface
- Instant predictions
- No technical knowledge required

**6.5 Discussion:**

- The model effectively detects contradictions in legal text
- Machine learning combined with NLP improves performance
- The system is efficient for real-time applications

**Limitations:**

- Complex legal sentences may reduce accuracy
- Limited dataset size may affect generalization
- Does not yet analyse full documents (only sentence pairs)

**6.6 Summary:**

This chapter demonstrated that the developed system performs effectively in detecting contradictions and relationships in legal text. The results show that NLP and machine learning techniques can be successfully applied to real-world legal analysis problems.

# Chapter 7 – Conclusion and Future Scope

### 7.1 Summary

The project titled "Legal Document Contradiction Summary Generator using Machine Learning and NLP" successfully demonstrates how artificial intelligence techniques can be applied to analyse and interpret legal text. The system is designed to identify contradictions, entailment, and neutral relationships between legal statements.

The implementation involves preprocessing legal text, converting it into numerical form using TF-IDF vectorization, and applying a machine learning model for classification. The developed system is capable of analysing sentence pairs and generating accurate predictions.

Additionally, a Streamlit-based web application is created to provide a simple and interactive interface for users to input legal statements and obtain real-time results. The system improves efficiency, reduces manual effort, and enhances accuracy in legal document analysis.

### 7.2 Applications

This project has several real-world applications in the legal and business domains:

- **Legal Document Review:** Detect contradictions in contracts and agreements
- **Compliance Checking:** Ensure consistency in policies and regulations
- **Law Firms:** Assist lawyers in reviewing lengthy documents
- **Contract Management Systems:** Improve reliability of agreements
- **Data Analytics:** Analyze patterns in legal text

### 7.3 Future Scope

The project can be further improved and extended in the following ways:

1. Advanced Machine Learning Models

- Implement models such as:
- Random Forest

- Support Vector Machines
- XGBoost

2. Deep Learning Techniques
   - Use advanced NLP models:
   - LSTM (Long Short-Term Memory)
   - BERT (Bidirectional Encoder Representations from Transformers)
3. Larger and Better Dataset
   - Use more real-world legal datasets
   - Improve model accuracy and generalization
4. Full Document Analysis
   - Extend system from sentence-level to full document analysis
   - Detect contradictions across entire contracts
5. Multilingual Support
   - Support multiple languages
   - Expand usability globally
6. Enhanced User Interface
   - Improve UI design
   - Add features like:
   - History tracking
   - Report generation
   - Visualization

**7.4 Final Conclusion:**

This project highlights the potential of machine learning and NLP in solving real-world legal challenges. The developed system provides a strong foundation for building more advanced legal AI tools and demonstrates how technology can significantly improve efficiency and accuracy in document analysis.

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